

## Scenario A: Correct Client Certificate

Goal: Verify that normal mTLS connections work.

| Field                  | Your Result                                                                   |
|------------------------|-------------------------------------------------------------------------------|
| Connection outcome     | <input checked="" type="checkbox"/> Succeeded <input type="checkbox"/> Failed |
| Error message (if any) | NONE                                                                          |
| Expected outcome       | Connection succeeds                                                           |
| Test result            | <input checked="" type="checkbox"/> PASS <input type="checkbox"/> FAIL        |

Analysis: Did the connection succeed as expected?

Yes, the connection was successful as expected. The client using a valid certificate was successfully able to complete the handshake and authenticate the broker.

Screenshot: (Paste your screenshot below)

```
kaylacolletti@Kaylas-MacBook-Pro hydroficient-project % python3 identity_tester.py --mode test-correct
=====
IDENTITY ATTACK SIMULATION
=====

SCENARIO A: Correct Client Certificate
=====
Testing connection with valid device-001 certificate...
/Users/kaylacolletti/Documents/IoT-Externship/hydroficient-project/identity_tester.py:275: DeprecationWarning: Callback API version 1 is deprecated, update to latest version
client = mqtt.Client(client_id="test-correct-cert", **MQTT_CLIENT_ARGS)

=====
TEST: Valid Client Certificate
=====
Expected: Connection succeeds
Actual: CONNECTION SUCCEEDED

>>> TEST PASSED - Connection succeeded as expected <<<
kaylacolletti@Kaylas-MacBook-Pro hydroficient-project %
```

## Scenario B: No Client Certificate

Goal: Verify that the broker rejects devices without certificates.

| Field                  | Your Result                                                                   |
|------------------------|-------------------------------------------------------------------------------|
| Connection outcome     | <input type="checkbox"/> Succeeded <input checked="" type="checkbox"/> Failed |
| Error message (if any) | Connection failed with rc=-1                                                  |
| Expected outcome       | Connection rejected                                                           |
| Test result            | <input type="checkbox"/> PASS <input checked="" type="checkbox"/> FAIL        |

Analysis: What specific error did you see? This error proves that a rogue device cannot connect just by having network access.

The error seen was “Connection failed with rc= -1”. This shows that the client was unable to establish a connection with the broker because it did not provide a client certificate.

Screenshot: (Paste your screenshot below)

```
''' TEST PROGRAM - CONNECTION SUCCESS OR FAILURE '''
kaylacolletti@Kaylas-MacBook-Pro hydroficient-project % python identity_tester.py --mode test-no-cert
zsh: command not found: python
kaylacolletti@Kaylas-MacBook-Pro hydroficient-project % python3 identity_tester.py --mode test-no-cert
=====
IDENTITY ATTACK SIMULATION
=====
[
]
=====
SCENARIO B: No Client Certificate
=====
Testing connection WITHOUT any client certificate...
/Users/kaylacolletti/Documents/IoT-Externship/hydroficient-project/identity_tester.py:314: DeprecationWarning: Callback API version 1 is deprecated, update to latest version
client = mqtt.Client(client_id="test-no-cert", **MQTT_CLIENT_ARGS)
=====
TEST: No Client Certificate
=====
Expected: Connection rejected
Actual: CONNECTION FAILED
Error: Connection failed with rc=-1
>>> TEST PASSED - Connection was correctly rejected <<<
kaylacolletti@Kaylas-MacBook-Pro hydroficient-project %
```

## Scenario C: Certificate from Wrong CA

Goal: Verify that only certificates signed by your CA are accepted.

| Field                  | Your Result                  |
|------------------------|------------------------------|
| Connection outcome     | [ ] Succeeded [X] Failed     |
| Error message (if any) | Connection failed with rc=-1 |
| Expected outcome       | Connection rejected          |
| Test result            | [X] PASS [ ] FAIL            |

Analysis: How does this differ from Scenario B? Both are rejected, but for different reasons.

The connection was rejected with the error of “Connection failed with rc=-1”. While this result is similar to Scenario B (no certificate), the reason is different. In this, the client did provide a certificate however it was signed by an untrusted (rogue) Certificate Authority. The broker rejected the connection because it could not verify the certificate against its trusted CA. This demonstrated that even if a device presents a certificate, it will be allowed to connect unless the certificate is issued by a trusted authority.

Screenshot: (Paste your screenshot below)

```
-----
/Users/kaylacolletti/Documents/IoT-Externship/hydroficient-project/identity_tester.py:314: DeprecationWarning: Callback API version 1 is deprecated, update to latest version
client = mqtt.Client(client_id="test-no-cert", **MQTT_CLIENT_ARGS)
=====
TEST: No Client Certificate
=====
Expected: Connection rejected
Actual: CONNECTION FAILED
Error: Connection failed with rc=-1

>>> TEST PASSED - Connection was correctly rejected <<<
kaylacolletti@Kaylas-MacBook-Pro hydroficient-project % python3 identity_tester.py --mode test-wrong-ca
=====
IDENTITY ATTACK SIMULATION
=====

SCENARIO C: Certificate from Wrong CA
-----
Generating rogue CA and client certificate...
Done. Rogue CA and device certificate created.
Testing connection with certificate signed by DIFFERENT CA...
/Users/kaylacolletti/Documents/IoT-Externship/hydroficient-project/identity_tester.py:359: DeprecationWarning: Callback API version 1 is deprecated, update to latest version
client = mqtt.Client(client_id="test-wrong-ca", **MQTT_CLIENT_ARGS)
=====
TEST: Wrong CA Certificate
=====
Expected: Connection rejected
Actual: CONNECTION FAILED
Error: Connection failed with rc=-1

>>> TEST PASSED - Connection was correctly rejected <<<
kaylacolletti@Kaylas-MacBook-Pro hydroficient-project % █
```

Scenario D: Expired Certificate

Goal: Verify that expired certificates are rejected.

| Field                  | Your Result                  |
|------------------------|------------------------------|
| Connection outcome     | [ ] Succeeded [ ] Failed     |
| Error message (if any) | Connection failed with rc=-1 |
| Expected outcome       | Connection rejected          |
| Test result            | [ ] PASS [ ] FAIL            |

Analysis: Why is certificate expiration important for IoT security? What happens if a device certificate never expires?

Certification expiration is very important because it limits how long devices credentials are valid for, this reduces the risk if they are compromised. In this case, the expired certificate was correctly rejected by the broker. If certificates never expire, a stolen or compromised certificate could be used indefinitely, creating a serious security risk.

Screenshot: (Paste your screenshot below)

```
-----
kaylacolletti@Kaylas-MacBook-Pro hydroficient-project % python3 identity_tester.py --mode test-expired
=====
IDENTITY ATTACK SIMULATION
=====

SCENARIO D: Expired Certificate
-----
Generating expired client certificate...
Done. Expired certificate created (expired yesterday).
Testing connection with EXPIRED client certificate...
/Users/kaylacolletti/Documents/IoT-Externship/hydroficient-project/identity_tester.py:412: DeprecationWarning: Callback API version 1 is deprecated, update to latest version
client = mqtt.Client(client_id="test-expired", **MQTT_CLIENT_ARGS)
=====
TEST: Expired Certificate
=====
Expected: Connection rejected
Actual: CONNECTION FAILED
Error: Connection failed with rc=-1

>>> TEST PASSED - Connection was correctly rejected <<<
kaylacolletti@Kaylas-MacBook-Pro hydroficient-project % █
```

## Summary Worksheet

Complete this summary of your identity attack simulations:

| Scenario        | What You Tried                      | Expected | Actual                     | Pass/Fail |
|-----------------|-------------------------------------|----------|----------------------------|-----------|
| A: Correct cert | Connect with valid device-001 cert  | Success  | Connection success         | PASS      |
| B: No cert      | Connect without client certificate  | Rejected | Connection failed (rc= -1) | PASS      |
| C: Wrong CA     | Connect with cert from different CA | Rejected | Connection failed (rc= -1) | PASS      |
| D: Expired      | Connect with expired certificate    | Rejected | Connection failed (rc= -1) | PASS      |